National 5 Engineering Science

Analogue Electronics

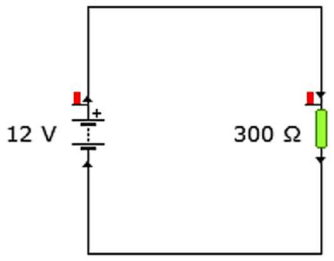
Section 1 — Attempt ALL questions

Total marks: 40

Name: _Marking Instructions_____ Class: _____ Date: _____

Question 1 (4 marks)

A resistor of 300 Ω is connected to a 12 V supply.



(a) Calculate the current flowing through the resistor. (2 marks)

V = IR

I = V / R = 12 / 300 = **0.04A**

(b) State the formula used. (1 mark)

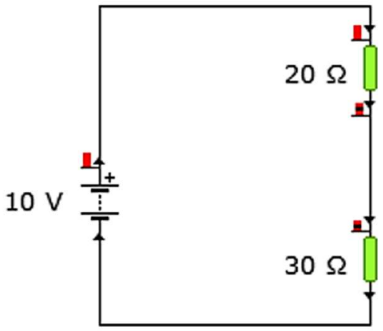
Ohm's Law, V = IR

(c) State the unit of current. (1 mark)

Amperes or Amps (A)

Question 2 (6 marks)

Two resistors of 20 Ω and 30 Ω are connected in series to a 10 V supply.



(a) Calculate the total resistance. (1 mark)

$R_{total} = R_1 + R_2$

$R_{total} = 20 + 30 =$ **50 Ω**

(b) Calculate the circuit current. (2 marks)

V = IR

I = V / R = 10 / 50 = **0.2A**

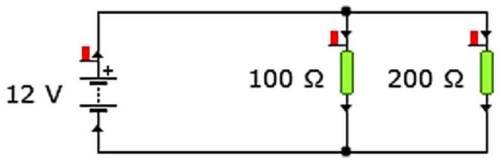
(c) Calculate the voltage across the 30 Ω resistor. (3 marks)

V = IR

$V_{30} = I \times R_{30} = 0.2 \times 30 =$ **6V**

Question 3 (6 marks)

Two resistors of 100 Ω and 200 Ω are connected in parallel to a 12 V supply.



(a) Calculate the total resistance. (3 marks)

$$R_{Total} = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$R_{Total} = \frac{100 \times 200}{100 + 200} = 66.6667 = \mathbf{66.7\Omega}$$

(b) Calculate the total circuit current. (2 marks)

V = IR

I = V / R = 12 / 66.67 = **0.18A**

(c) State one characteristic of a parallel circuit. (1 mark)

The voltage is the same as the supply voltage across each branch.

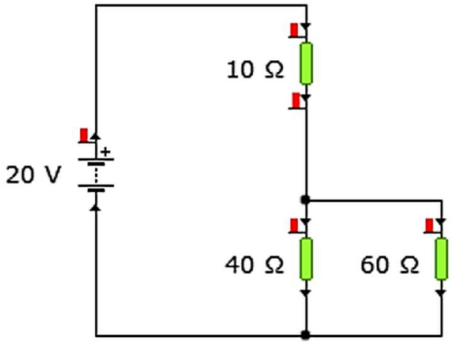
Question 4 (5 marks)

A 10 Ω resistor is connected in series with two resistors (40 Ω and 60 Ω) in parallel.

The supply voltage is 20 V.

(a) Calculate the total resistance of the parallel section. (2 marks)

$$R_{parallel} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{40 \times 60}{40 + 60} = \mathbf{24\Omega}$$



(b) Calculate the total circuit resistance. (1 mark)

$R_{total} = R_1 + R_{parallel}$

$R_{total} = 10 + 24 =$ **34 Ω**

(c) Calculate the circuit current. (2 marks)

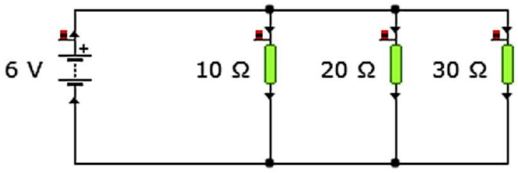
V = IR

I = V / R = 20 / 34 = 0.588235= **0.59A**

Question 5 (5 marks)

Three resistors of 10 Ω, 20 Ω and 30 Ω are connected in parallel to a 6 V supply.

(a) Calculate the total resistance. (3 marks)



$$\frac{1}{R_{Total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{Total}} = \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

$$\frac{1}{R_{Total}} = 0.183333$$

$$R_{Total} = 1 \div 0.183333 = 5.454545 = \mathbf{5.45\Omega}$$

(b) Calculate the current through the 10 Ω resistor. (2 marks)

V = IR

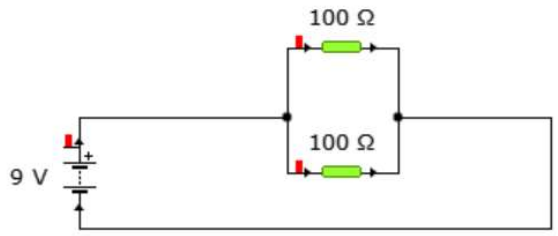
$I_{10} = V / R_{10}$

$I_{10} = 6 / 10 =$ **0.6A**

Question 6 (4 marks)

A student requires a resistance of 50 Ω but only has two 100 Ω resistors.

(a) Draw a circuit to achieve 50 Ω. (2 marks)



[Space for diagram]

(b) Prove by calculation that the total resistance is 50 Ω. (2 marks)

$$\frac{1}{R_{Total}} = \frac{1}{R_1} + \frac{1}{R_2}$$

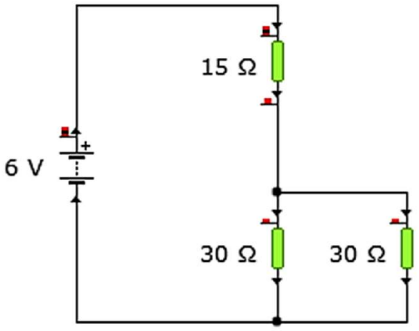
$$\frac{1}{R_{Total}} = \frac{1}{100} + \frac{1}{100}$$

$$\frac{1}{R_{Total}} = 0.02$$

$$R_{Total} = \frac{1}{0.02} = \mathbf{50\Omega}$$

Question 7 (5 marks)

A 6 V supply feeds a circuit where a 15 Ω resistor is connected in series with two 30 Ω resistors in parallel.



(a) Calculate the total resistance. (2 marks)

$$\frac{1}{R_{parallel}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{parallel}} = \frac{1}{30} + \frac{1}{30} = 0.066666\Omega$$

$$R_{parallel} = \frac{1}{0.066666} = 15\Omega$$

$$R_{Total} = R_{parallel} + 15 = 15 + 15 = \mathbf{30\Omega}$$

(b) Calculate the circuit current. (1 mark)

V = IR

I = V/ R

I = 6 / 30 = **0.2A**

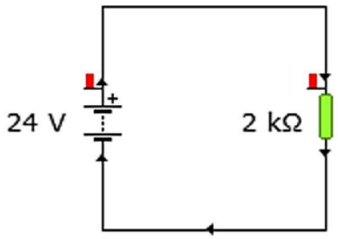
(c) Calculate the voltage across the parallel section. (2 marks)

$V_{parallel} = IR_{parallel}$

$V_{parallel}$ = 0.2 x 15 = **3V**

Question 8 (5 marks)

A 24 V supply is connected to a 2 kΩ resistor.



(a) Calculate the current in milliamps (mA). (3 marks)

V = IR

I = V/ R

I = 24 / 2000 = **0.012A**

(b) The resistor is replaced with a 1 kΩ resistor.

State what happens to the current and explain why. (2 marks)

The current increases as the resistance reduces. To be more specific, because the resistance is halved, the current will therefore double.

Proof:

I = V/ R

I = 24 / 1000 = **0.024A**